

Krishnaveni Parvataneni

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Education

Massachusetts Institute of Technology

Artificial Intelligence and Decision Making

Cambridge, MA

Aug 2023 - May 2027

Relevant Coursework: Fundamentals of Programming, Mathematics for Computer Science, Organic Chemistry I, Linear Algebra, Introduction to Algorithms, Introduction to Machine Learning, Signal Processing

Experience

Precision in Medical Imaging Lab, Massachusetts General Hospital (MGH)

Boston, MA

Researcher

Dec 2023 - Present

- Mentor: Dr. Dania Daye, MD, PhD
- Implemented a system utilizing LLMs to automate abscess drain management for patients of the MGH radiology department.
- Developing machine learning-based risk assessment models for predicting 30-day rehospitalization of hemodialysis patients following IR procedures.

Graybiel Lab, McGovern Institute for Brain and Cognitive Sciences

Cambridge, MA

Undergraduate Researcher

Jul 2023 - Present

- Mentor: Dr. Iakovos Lazaridis, PhD, Dr. Tomoko Yoshida, MD
- Employing data analysis techniques and machine learning to analyze neuronal recording signals and movement kinematics during decision-making behaviors in freely moving mice.
- Conducting protein expression analysis following virus injection in mouse striatum and substantia nigra (immunohistochemistry).

Traverso Lab, MIT Department of Mechanical Engineering

Cambridge, MA

Undergraduate Researcher

Jul 2023 - Present

- Mentor: Dr. Troy (Ziliang) Kang, PhD
- Creating models in SOLIDWORKS and Fusion 360 for soft robotics molds, testing their functionality for gastrointestinal health monitoring.
- Designing a star device for oxalate oxidase delivery to treat hyperoxaluria.
- Testing rapid and delayed release capsules in diverse bodily conditions.

Edelman Lab, MIT-Harvard Health Sciences and Technology

Cambridge, MA

Undergraduate Researcher

Jan 2023 - Present

- Mentor: Kimberly K. Lamberti
- Utilizing Computer Vision (Python) for classifying swine echocardiography images to complement mechanical pacemaker data.

Harvard T.H. Chan School Of Public Health

Boston, MA

Teaching Assistant

Jul 2023 - Aug 2024

- Assisted in teaching BST.209, a machine learning/data science course in R for doctors at Harvard Medical Campus. Collaborated with a team of doctors to develop a project aimed at predicting the disposition of patients presenting to the ED with Coronary Heart Failure, with a focus on bias reduction.

MIT Laboratory for Computational Physiology

Cambridge, MA

Research Intern

Jun 2022 - Feb 2024

- Mentor: Dr. Leo A. Celi, MD
- Analyzing patient-based medical app start-up failures to enhance strategies for medical startups.
- Investigating bias introduced by Shortcut learning in medical image-based classification models.

Kellis Lab, MIT CSAIL

Undergraduate Researcher

Cambridge, MA

Aug 2023 - Feb 2024

- Mentor: Dr. Xikun Han, PhD
- Utilizing Machine Learning to develop a pipeline that employs single-cell genome sequencing to explore the impact of structural variations and mobile elements on the risk of neuropsychiatric diseases in Python and R

CognoTrain Inc

Front-End Engineering Intern

Remote

Feb 2021 - Aug 2023

- Mentor: Sahar Jahanikia
- Front-End Engineer at Cognotrain (www.cognotrain.com), designing activities, screens, and views in Flutter based on patient data using CBCT/CBCR to personalize care for Alzheimer's and Dementia patients

Selected Publications

I. Lazaridis, J.R. Crittenden, G. Ahn, K. Hirokane, T. Yoshida, A. Mahar, V. Skara, K. Meletis, J.H. Loftus, **K. Parvataneni**, J.T. Ting, E. Hueske, A. Matsushima and A.M. Graybiel, "Striosomes Target Nigral Dopamine-Containing Neurons via Direct-D1 and Indirect-D2 Pathways Paralleling Classic Direct-Indirect Basal Ganglia Systems," *bioRxiv*, doi: 10.1101/2024.06.01.596922.

W. Xu, J. Zhao, **K. Parvataneni**, V. Zhang, S. Lin and H. Wang, "Hybrid AI Framework for Remote Patient Heart Failure Risk Prediction," 2022 Global Reliability and Prognostics and Health Management (PHM-Yantai), Yantai, China, 2022, pp. 1-8, doi: 10.1109/PHM-Yantai55411.2022.9941764.

K. Parvataneni and S. Zaidi, "Exploring the Vibrational and Rotational Temperatures of a DBD Plasma Jet for Wound Healing," 2022 IEEE-EMBS Conference on Biomedical Engineering and Sciences (IECBES), Kuala Lumpur, Malaysia, 2022, pp. 218-222, doi: 10.1109/IECBES54088.2022.10079657.

Skills & Interests

Technical: Python, R, Java, C, C++, Q#, SQL, Flutter, JavaScript, HTML/CSS, SolidWorks, FUSION360
Language: Telugu (bilingual proficiency), Spanish (working proficiency), Italian (elementary proficiency)
Laboratory: immunohistochemical staining, ELISA, PCR, brain sectioning, animal handling (pigs/mice)